List of Courses for B.E. / M.Sc. / B.Pharm. Programmes:

The list of Discipline Core Courses and Discipline Electives for all the first degree programmes is given below. To complete the requirements of Humanities electives, a student can take courses which are normally listed under Languages and Literature, History and Philosophy, Political and social Sciences, Fine arts and Professional Arts which have also been mentioned in subsequent paragraphs.

ARCHITECTURAL AND URBAN ENGINEERING

CORE COURSES

Course No	Course Title	L	P	U
AUE F211	Basic Design for Visualization			3
AUE F212	Structural Mechanics	3	0	3
AUE F213	Fluid Mechanics and Applications	3	0	3
AUE F241	Architectural Design Studio I			3
AUE F242	Building Construction & Technology			3
AUE F243	Construction Economics	3	0	3
AUE F311	Architectural Design Studio II			3
AUE F312	Geotechnical Design	3	0	3
AUE F313	Design of Reinforced Concrete structures	3	0	3
AUE F314	Highway Planning, Analysis and Design	3	1	4
AUE F341	Urban Planning and Sustainable Communities			3
AUE F342	Building Acoustics and Lighting Design			2
AUE F343	Design of steel structures	3	0	3
AUE F344	Directed Research in Architecture – I			1
AUE F345	Directed Research in Architecture – II			4
CE F230	Civil Engineering Materials	3	2	4

DISCIPLINE ELECTIVE COURSES

Track - 1: Environment and sustainable design

AUE F361	Landscape Architecture	3	0	3
AUE F362	Building Services	3	0	3
AUE F421	Renewable Energy Systems in Buildings	3	0	3
CE F323	Introduction to Environmental Engineering	3	0	3
CE F423	Green Building and Energy Conservation	3	0	3
CE F434	Environmental Impact Assessment	3	0	3
ME F433	Solar Thermal Process Engineering	3	1	4

Track - 2: Urban studies

AUE F422	Urban Mass Transit Planning, Operations and Management	3	1	4
CE F321	Engineering Hydrology	3	0	3
CE F342	Water & Waste Water Treatment	3	1	4
CE F420	Introduction to Bridge Engineering	3	0	3
CE F422	Urban Hydrology	2	1	3
CE F425	Airport, Railways and Waterways	3	0	3
CE F428	Earthquake Resistant Design and Construction	3	0	3
ME G513	Heating and Cooling of Buildings	3	2	5

Track - 3: Emerging Technologies

AUE F423	Machine Learning in Design Optimization	3	0	3
AUE F425	Building Information Modeling (BIM)			3
CE F416	Computer Applications in Civil Engineering	3	1	4
CE F417	Applications of Artificial Intelligence in Civil Engineering	3	0	3
CE F431	Principles of Geographical Information Systems	3	1	4
CE F433	Remote Sensing and Image Processing	3	1	4

Track - 4: Construction & Finance

AUE F424	Construction Management	3	1	4
BITS F469	Financing Infrastructure projects	3	0	3
CE F427	System Modeling and Analysis	3	0	3

BIOTECHNOLOGY

CORE COURSES

		L	P	U
BIOT F211	Biological Chemistry	3	0	3
BIOT F212	Microbiology	3	1	4
BIOT F213	Cell biology	3	0	3
BIOT F215	Biophysics	3	0	3
BIOT F241	Genetic Engineering Techniques	1	3	4
BIOT F243	Genetics	3	0	3
BIOT F244	Instrumental Methods of Analysis	1	3	4
BIOT F245	Intro to Environmental Biotechnology	3	0	3
BIOT F311	Recombinant DNA Technology	3	0	3
BIOT F314	Industrial Microbiology and Bioprocess Engineering	2	2	4
BIOT F342	Immunology	3	0	3
BIOT F343	Experiments in Biotechnology	0	3	3
BIOT F344	Downstream Processing	2	1	3

DISCIPLINE ELECTIVE COURSES

BIOT F242	Introduction to Bioinformatics	3	0	3
BIOT F345	Proteomics	3	0	3
BIOT F346	Genomics	3	0	3
BIOT F347	Immunotechnology	3	0	3
BIOT F352	Cell and Tissue Culture Technology	3	0	3

BIOT F413	Molecular Biology of the Cell	3	0	3
BIOT F416	Introduction to Pharmaceutical Biotechnology	3	0	3
BIOT F417	Biomolecular Modeling	3	0	3
BIOT F420	Introduction to Plant Biotechnology	3	0	3
BIOT F422	Nanobiotechnology	3	0	3
BIOT F423	Drug design and delivery	3	0	3
BIOT F424	Food Biotechnology	3	0	3
BITS F467	Bioethics and Biosafety	3	0	3

BIOTECHNOLOGY WITH SPECIALIZATION IN APPLIED MOLECULAR BIOLOGY

CORE COURSES		**L**	**P**	**U**
BIOT F211	Biological Chemistry	3	0	3
BIOT F212	Microbiology	3	1	4
BIOT F213	Cell biology	3	0	3
BIOT F215	Biophysics	3	0	3
BIOT F241	Genetic Engineering Techniques	1	3	4
BIOT F243	Genetics	3	0	3
BIOT F244	Instrumental Methods of Analysis	1	3	4
BIOT F245	Intro to Environmental Biotechnology	3	0	3
BIOT F311	Recombinant DNA Technology	3	0	3
BIOT F314	Industrial Microbiology and Bioprocess Engineering	2	2	4
BIOT F342	Immunology	3	0	3
BIOT F343	Experiments in Biotechnology	0	3	3
BIOT F344	Downstream Processing	2	1	3

DISCIPLINE ELECTIVE COURSES

BIOT F345	Proteomics	3	0	3
BIOT F346	Genomics	3	0	3
BIOT F413*	Molecular Biology of the Cell*	3	0	3
BIOT F492*	Applied Molecular Biology Project*			3
BIOT F242	Introduction to Bioinformatics	3	0	3
BIOT F347	Immunotechnology	3	0	3
BIOT F416	Introduction to Pharmaceutical Biotechnology	3	0	3
BIOT F420	Introduction to Plant Biotechnology	3	0	3

* Compulsory discipline elective courses out of the discipline electives requirement

CHEMICAL ENGINEERING

CORE COURSES		**L**	**P**	**U**
CHE F211	Chemical Process Calculations	3	0	3
CHE F212	Fluid Mechanics	3	0	3
CHE F213	Chemical Engineering Thermodynamics	3	0	3
CHE F214	Engineering Chemistry	3	0	3
CHE F241	Heat Transfer	3	0	3
CHE F242	Numerical Methods for Chemical Engineers	3	0	3
CHE F243	Material Science and Engg.	3	0	3
CHE F244	Separation Processes I	3	0	3
CHE F311	Kinetics and Reactor Design	3	0	3
CHE F312	Chemical Engineering Lab I	0	3	3
CHE F313	Separation Processes II	3	0	3
CHE F314	Process Design Principles I	3	0	3
CHE F341	Chemical Engineering Lab II	0	3	3
CHE F342	Process Dynamics and Control	3	0	3
CHE F343	Process Design Principles II	3	0	3

DISCIPLINE ELECTIVE COURSES

BIO G671	Bioconversion Technology	3	2	5
BIOT F245	Introduction to Environmental Biotechnology	3	0	3
BIOT F344	Downstream processing	2	1	3
BITS F415	Introduction to MEMS	3	1	4
BITS F416	Introduction to Nanoscience	3	0	3
BITS F417	Microfluidics and its application			4
BITS F418	Introduction to Biomedical Engineering	3	1	4
BITS F429	Nanotechnology for Renewable Energy and Environment	3	1	4
CHE F315	Machine Learning for Chemical Engineers	3	0	3
CHE F316	Sustainable Energy Systems	3	0	3
CHE F317	Energy Systems Engineering	3	0	3
CHE F411	Environmental Pollution Control	3	0	3
CHE F412	Process Equipment Design	3	0	3
CHE F413	Process Plant Safety	3	0	3
CHE F414	Transport Phenomena	3	0	3
CHE F415	Molecular and Statistical Thermodynamics	3	0	3
CHE F416	Process Plant Design Project I	-	-	3
CHE F417	Process Plant Design Project II	-	-	3
CHE F418	Modelling and Simulation in Chemical Engineering	3	0	3
CHE F419	Chemical Process Technology	3	0	3
CHE F421	Bio-chemical Engineering	3	0	3
CHE F422	Petroleum Refining Technology	3	0	3
CHE F423	Membrane Science and Engineering	3	0	3
CHE F424	Rheology of Complex Fluids	3	0	3
CHE F425	Environment Management Systems	3	0	3
CHE F433	Corrosion Engineering	3	0	3
CHE F471	Advanced Process Control	3	0	3
CHE F497	Atomic and Molecular Simulations			3

CHE F498	Colloids and Interface Engineering			3
CHE G511	Fluidization Engineering	3	1	4
CHE G512	Petroleum Refining and Petrochemicals	3	1	4
CHE G513	Environmental Management Systems	3	2	5
CHE G522	Polymer Technology	3	1	4
CHE G523	Mathematical Methods in Chemical Engineering			5
CHE G524	Introduction to Multiphase flow	3	1	4
CHE G526	Nuclear Engineering	3	1	4
CHE G527	Energy Conservation and Management	3	1	4
CHE G528	Introduction to Nanoscience & Technology	3	1	4
CHE G529	Pulp & Paper Technology	3	1	4
CHE G532	Alternate Energy Resources	3	1	4
CHE G533	Petroleum Product Characterization	3	2	5
CHE G551	Advanced Separation Technology	3	2	5
CHE G552	Advanced Transport Phenomena			5
CHE G554	Computational Fluid Dynamics			4
CHE G556	Electrochemical Engineering	3	1	4
CHE G557	Energy Systems Engineering			4
CHE G558	Chemical Process Optimization			4
CHE G568	Modeling and Simulation in Petroleum Refining			4
CHE G613	Advanced Mass Transfer	3	2	5
CHE G614	Advanced Heat Transfer	3	2	5
CHE G616	Petroleum Reservoir Engineering			5
CHE G617	Petroleum Refinery Engg.	3	2	5
CHE G618	Petroleum Downstream Processing	3	2	5
CHE G619	Process Intensification	3	2	5
CHE G620	Energy Integration Analysis	3	1	4
CHE G622	Advanced Chemical Engineering Thermodynamics			5
CHE G641	Reaction Engineering			5
CHEM F325	Polymer Chemistry	3	0	3
ME F323	Energy Storage Technologies	3	0	3
MST F332	Materials Processing	3	0	3
MST F334	Materials for Catalytic Applications	3	0	3
MST F339	Polymer Materials	3	0	3
MST G521	Material Characterization Techniques	3	2	5

CHEMICAL ENGINEERING WITH SPECIALIZATION IN ENERGY, ENVIRONMENT, AND SUSTAINABILITY

CORE COURSES		L	P	U
CHE F211	Chemical Process Calculations	3	0	3
CHE F212	Fluid Mechanics	3	0	3
CHE F213	Chemical Engineering Thermodynamics	3	0	3
CHE F214	Engineering Chemistry	3	0	3
CHE F241	Heat Transfer	3	0	3
CHE F242	Numerical Methods for Chemical Engineers	3	0	3
CHE F243	Material Science and Engineering	3	0	3
CHE F244	Separation Processes I	3	0	3
CHE F311	Kinetics and Reactor Design	3	0	3
CHE F312	Chemical Engineering Lab I	0	3	3
CHE F313	Separation Processes II	3	0	3
CHE F314	Process Design Principles I	3	0	3
CHE F341	Chemical Engineering Lab II	0	3	3
CHE F342	Process Dynamics and Control	3	0	3
CHE F343	Process Design Principles II	3	0	3

DISCIPLINE ELECTIVE COURSES

Pool-1 (Three Courses from pool-1)

CHE F316	Sustainable Energy Systems	3	0	3
CHE F317	Energy Systems Engineering	3	0	3
or				
CHE G557	or Energy Systems Engineering	0	0	4
CHE F425	Environmental Management Systems	3	0	3
or				
CHE G513	or Environmental Management Systems	3	2	5

Pool-2 (Any Two Courses from pool-2)

BITS F217	Environment, Development and Climate Change	3	0	3
CHE F411	Environmental Pollution Control	3	0	3
ECON F353	Energy Economics and Policy	3	0	3
ME F323	Energy Storage Technologies	3	0	3
CHE F366	Laboratory Project			3
or				
CHE F376	Design Project			
or				
CHE F491	Special Project			
HSS F381	Environmental Sustainability Ventures	3	0	3

CIVIL ENGINEERING

CORE COURSES		L	P	U
CE F211	Mechanics of Solids	3	0	3
CE F213	Surveying	3	1	4
CE F230	Civil Engineering Materials			4

CE F231	Fluid Mechanics			3
CE F241	Analysis of Structures	3	0	3
CE F242	Construction Planning and Technology	3	0	3
CE F243	Soil Mechanics	3	1	4
CE F244	Highway Engineering	3	1	4
CE F312	Hydraulics Engineering	3	1	4
CE F313	Foundation Engineering	3	0	3
CE F320	Design of Reinforced Concrete Structures			3
CE F321	Engineering Hydrology			3
CE F342	Water & Waste Water Treatment	3	1	4
CE F343	Design of Steel Structures	3	0	3
DISCIPLINE ELECTIVE COURSES				
BITS F313	Multicriterion Decision Making in Engg. and Management	3	0	3
CE F323	Introduction to Environmental Engineering	3	0	3
CE F324	Numerical Analysis	3	0	3
CE F325	Fundamentals of Rock Mechanics	3	0	3
CE F331	In-situ Testing Methods in Geotechnical Engineering	3	0	3
CE F345	Computational Geomechanics	3	0	3
CE F411	Operation Research for Engineers	3	0	3
CE F412	Disaster Management	3	0	3
CE F413	Advanced Structural Design	3	0	3
CE F415	Design of Prestressed Concrete Structure	3	0	3
CE F416	Computer Applications in Civil Engineering	3	1	4
CE F417	Applications of Artificial Intelligence in Civil Engg.	3	0	3
CE F419	Geotechnical Earthquake Engg. and Machine Foundation	3	0	3
CE F420	Introduction to Bridge Engineering	3	0	3
CE F421	Analysis and Design of FRP Reinforced Concrete Structures	3	0	3
CE F422	Urban Hydrology	2	1	3
CE F423	Green Buildings and Energy Conservation	3	0	3
CE F425	Airport, Railways and Waterways	3	0	3
CE F426	Geosynthetics and Reinforced Soil Structure	3	0	3
CE F427	System Modeling and Analysis	3	0	3
CE F428	Earthquake Resistant Design and Construction	3	0	3
CE F429	Design of Foundation Systems	3	0	3
CE F430	Design of Advanced Concrete Structures	3	0	3
CE F431	Principles of Geographical Information Systems	3	1	4
CE F432	Structural Dynamics	3	0	3
CE F433	Remote Sensing and Image Processing	3	1	4
CE F434	Environmental Impact Assessment	3	0	3
CE F435	Introduction to Finite Element Methods	3	0	3

COMPUTER SCIENCE

CORE COURSES		**L**	**P**	**U**
CS F211	Data Structures & Algorithms	3	1	4
CS F212	Database Systems	3	1	4
CS F213	Object Oriented Programming	3	1	4
CS F214	Logic in Computer Science	3	0	3
CS F215	Digital Design	3	1	4
CS F222	Discrete Structures for Computer Science	3	0	3
CS F241	Microprocessors & Interfacing	3	1	4
CS F301	Principles of Programming Languages	2	0	2
CS F303	Computer Networks	3	1	4
CS F342	Computer Architecture	3	1	4
CS F351	Theory of Computation	3	0	3
CS F363	Compiler Construction	2	1	3
CS F364	Design & Analysis of Algorithms	3	0	3
CS F372	Operating Systems	3	0	3
DISCIPLINE ELECTIVE COURSES				
BITS F311	Image Processing	3	0	3
BITS F312	Neural Networks and Fuzzy Logic	3	0	3
BITS F343	Fuzzy Logic and Applications	3	0	3
BITS F364	Human – Computer Interaction	3	0	3
BITS F386	Quantum Information and Computation	3	0	3
BITS F452	Blockchain Technology	3	0	3
BITS F453	Computational Learning Theory	3	0	3
BITS F454	Bio-Inspired Intelligence: Algorithms and Applications	3	0	3
BITS F459	Computer Vision	3	1	4
BITS F463	Cryptography	3	0	3
BITS F464	Machine Learning	3	0	3
BITS F465	Enterprise Computing	3	1	4
BITS F466	Service Oriented Computing	3	1	4
BITS F471	Introduction to Large Language Models	3	0	3
CS F314	Software Development for Portable Devices	2	1	3

CS F315	Information and Communication Technologies and Development	3	0	3
CS F316	Quantum Architecture and Programming	3	0	3
CS F317	Reinforcement Learning	3	0	3
CS F320	Foundations of Data Science	3	0	3
CS F321	System Security	3	0	3
CS F322	Knowledge Graphs-Foundations and Applications	3	0	3
CS F401	Multimedia Computing	3	0	3
CS F402	Computational Geometry	3	0	3
CS F407	Artificial Intelligence	3	0	3
CS F413	Internetworking Technologies	3	0	3
CS F415	Data Mining	3	0	3
CS F422	Parallel Computing	3	0	3
CS F424	Software for Embedded Systems	3	1	4
CS F425	Deep Learning	3	0	3
CS F426	Graph Mining	3	1	4
CS F427	Performance Analysis of Computer Networks	3	0	3
CS F428	Special topic in Computer Science	1	0	1
CS F429	Natural Language Processing	3	0	3
CS F430	Approximation Algorithms	3	0	3
CS F431	Combinatorial Optimization	3	0	3
CS F432	Brain-inspired Deep Learning	3	0	3
CS F433	Computational Neuroscience	3	0	3
CS F434	Data science for Healthcare			3*
CS F435	Cyber Security Analytics and Forensics	3	1	4
CS F436	Cyber Physical Systems and Security	3	1	4
CS F437	Generative Artificial Intelligence	3	0	3
CS F441	Selected Topics from Computer Science	-	-	3
CS F444	Real Time Systems	3	0	3
CS F446	Data Storage Technologies and Networks	3	0	3
CS F468	Information Security Project	0	3	3
CS F469	Information Retrieval	3	0	3
CS G513	Network Security	3	1	4
CS G514/ SS G514	Object Oriented Analysis and Design	2	2	4
CS G519	Social Media Analytics	3	1	4
CS G520	Advanced Data Mining	3	1	4
CS G527	Cloud Computing			5
CS G557	Distributed Computing			5*
IS F311	Computer Graphics	3	0	3
IS F341	Software Engineering	3	1	4
IS F462	Network Programming	3	0	3
MATH F231	Number Theory	3	0	3
MATH F421	Combinatorial Mathematics	3	0	3
MATH F441	Discrete Mathematical Structures	3	0	3

ELECTRICAL AND ELECTRONICS ENGINEERING

CORE COURSES

EEE F211	Electrical Machines	3	1	4
EEE F212	Electromagnetic Theory	3	0	3
EEE F214	Electronic Devices	3	0	3
EEE F215	Digital Design	3	1	4
EEE F241	Microprocessors and interfacing	3	1	4
EEE F242	Control Systems	3	0	3
EEE F243	Signals & Systems	3	0	3
EEE F244	Microelectronic Circuits	3	0	3
EEE F311	Communication Systems	3	1	4
EEE F312	Power Systems	3	0	3
EEE F313	Analog & Digital VLSI Design	3	0	3
EEE F341	Analog Electronics	3	1	4
EEE F342	Power Electronics	3	1	4
MATH F212	Optimization	3	0	3
OR	OR			
ME F344	Engineering Optimization	2	0	2

DISCIPLINE ELECTIVE COURSES

BITS F312	Neural Networks and Fuzzy Logic	3	0	3
BITS F415	Introduction To MEMS	3	1	4
CS F213	Object Oriented Programming	3	1	4
CS F342	Computer Architecture	3	1	4
CS F372	Operating Systems	3	0	3
CS F451	Combinatorial Mathematics	3	0	3
CS G553	Reconfigurable Computing			5
ECE F312	EM Fields and Microwave Engineering Laboratory	0	1	1
ECE F314	EM Fields and Microwave Engineering	3	0	3
ECE F331	Microwave Engineering	3	1	4
ECE F343	Communication Networks	3	0	3
EEE F216	Electronic Devices Simulation Laboratory	0	2	2
EEE F245	Control System Laboratory	0	1	1
EEE F246	Electrical and Electronic Circuits Laboratory	0	2	2
EEE F345	Power Apparatus & Networks	3	0	3
EEE F346	Data Communication Networks	2	0	2
EEE F318	FPGA Based System Design	1	2	3
EEE F411	Internet of Things	3	1	4
EEE F414	Telecommunication Switching Systems & Networks	3	0	3
EEE F416	Digital Communication	3	0	3
EEE F417	Computer Based Control System	3	0	3

EEE F418	Modern Communication Technologies	3	0	3
EEE F419	Flexible and Stretchable Electronics	3	1	4
EEE F420	Biomedical Signal Processing	3	1	4
EEE F422	Modern Control Systems	3	0	3
EEE F423	Electronic Material Design and Simulations	1	2	3
OR	OR			
ECE F423	Electronic Material Design and Simulations	1	2	3
OR	OR			
INSTR F423	Electronic Material Design and Simulations	1	2	3
EEE F424	Smart Grid for Sustainable Energy	3	0	3
EEE F425	Power System Analysis and control	3	0	3
EEE F426	Fiber Optics & Optoelectronics	3	0	3
EEE F427	Electric Power Utilization and Illumination	3	0	3
EEE F428	Energy Storage Systems	3	0	3
EEE F429	Smart Materials and Applications	3	1	4
EEE F430	Green Communications and Networks	3	0	3
EEE F431	Mobile Telecommunication Networks	3	0	3
EEE F432	Medical Instrumentation	3	0	3
EEE F433	Electromagnetic Fields & Waves	3	0	3
EEE F434	Digital Signal Processing	3	1	4
EEE F435	Digital Image Processing	3	0	3
EEE F436	Electromagnetic Compatibility	3	1	4
EEE F437	Semiconductor Fabrication Technology	3	1	4
EEE F438	Electric Vehicles	3	0	3
EEE F462	Advanced Power Systems	3	0	3
EEE F472	Satellite Communication	3	0	3
EEE F473	Wind Electrical Systems	3	0	3
EEE F474	Antenna Theory and Design	3	1	4
EEE F475	Special Electrical Machines	3	1	4
EEE F476	Switchgear and Protection	3	1	4
EEE F477	Modelling of Field-Effect NanoDevices	3	0	3
EEE F478	Power Systems Laboratory	0	2	2
EEE G512	Embedded System Design	3	1	4
EEE G513	Machine Learning for Electronics Engineers			4*
EEE G626	Hardware Software Co-Design			4

ELECTRONICS AND COMMUNICATION ENGINEERING

CORE COURSES		L	P	U
ECE F211	Electrical Machines	3	1	4
ECE F212	Electromagnetic Theory	3	0	3
ECE F214	Electronic Devices	3	0	3
ECE F215	Digital Design	3	1	4
ECE F241	Microprocessors and interfacing	3	1	4
ECE F242	Control Systems	3	0	3
ECE F243	Signals & Systems	3	0	3
ECE F244	Microelectronic Circuits	3	0	3
ECE F311	Communication Systems	3	1	4
ECE F331	Microwave Engineering	3	1	4
ECE F341	Analog Electronics	3	1	4
ECE F343	Communication Networks	3	0	3
ECE F344	Information Theory & Coding	3	0	3
ECE F434	Digital Signal Processing	3	1	4

DISCIPLINE ELECTIVE COURSES		L	P	U
BITS F415	Introduction to MEMS	3	1	4
BITS F463	Cryptography	3	0	3
CS F213	Object Oriented Programming	3	1	4
CS F342	Computer Architecture	3	1	4
CS F372	Operating Systems	3	0	3
CS F451	Combinatorial Mathematics	3	0	3
CS G553	Reconfigurable Computing			5
ECE F216	Electronic Devices Simulation Laboratory	0	2	2
ECE F312	EM Fields and Microwave Engineering Laboratory	0	1	1
ECE F414	Telecommunication Switching Systems & Networks	3	0	3
ECE F416	Digital Communication	3	0	3
ECE F418	Modern Communication Technologies	3	0	3
ECE F424	Smart Grid for Sustainable Energy	3	0	3
ECE F428	Energy Storage Systems	3	0	3
ECE F431	Mobile Telecommunication Networks	3	0	3
ECE F472	Satellite Communication	3	0	3
EEE F245	Control System Laboratory	0	1	1
EEE F246	Electrical and Electronic Circuits Laboratory	0	2	2
EEE F313	Analog & Digital VLSI Design	3	0	3
EEE F345	Power Apparatus & Networks	3	0	3
EEE F346	Data Communication Networks	2	0	2
EEE F318	FPGA Based System Design	1	2	3
EEE F411	Internet of Things	3	1	4
EEE F417	Computer Based Control System	3	0	3

EEE F419	Flexible and Stretchable Electronics	3	1	4
EEE F420	Biomedical Signal Processing	3	1	4
EEE F422	Modern Control Systems	3	0	3
EEE F423	Electronic Material Design and Simulations	1	2	3
OR	OR			
ECE F423	Electronic Material Design and Simulations	1	2	3
OR	OR			
INSTR F423	Electronic Material Design and Simulations	1	2	3
EEE F426	Fiber Optics and Optoelectronics	3	0	3
EEE F429	Smart Materials and Applications	3	1	4
EEE F430	Green Communications and Networks	3	0	3
EEE F432	Medical Instrumentation	3	0	3
EEE F435	Digital Image Processing	3	0	3
EEE F436	Electromagnetic Compatibility	3	1	4
EEE F437	Semiconductor Fabrication Technology	3	1	4
EEE F438	Electric Vehicles	3	0	3
EEE F474	Antenna Theory and Design	3	1	4
EEE F475	Special Electrical Machines	3	1	4
EEE F476	Switchgear and Protection	3	1	4
EEE F477	Modelling of Field-Effect Nano Devices	3	0	3
EEE F478	Power Systems Laboratory	0	2	2
EEE G512	Embedded System Design	3	1	4
EEE G513	Machine Learning for Electronics Engineers			4*
EEE G626	Hardware Software Co-Design			4
INSTR F412	Analysis Instrumentation	3	0	3

ELECTRONICS AND COMPUTER ENGINEERING

CORE COURSES		L	P	U
ECOM F211	Data Structures and Algorithms	3	1	4
ECOM F213	Object Oriented Programming	3	1	4
ECOM F214	Electronic Devices	3	0	3
ECOM F215	Digital Design	3	1	4
ECOM F222	Discrete Structures for Computer Science	3	0	3
ECOM F241	Microprocessors and Interfacing	3	1	4
ECOM F242	Control Systems	3	0	3
ECOM F243	Signals & Systems	3	0	3
ECOM F244	Microelectronic Circuits	3	0	3
ECOM F313	Analog & Digital VLSI Design	3	0	3
ECOM F321	Real Time Operating Systems	3	1	4
ECOM F342	Computer Architecture	3	1	4
ECOM F343	Communication Networks	3	0	3
ECOM F462	Network Programming	3	0	3

DISCIPLINE ELECTIVE COURSES		L	P	U
BITS F312	Neural Networks and Fuzzy Logic	3	0	3
BITS F327	Artificial Intelligence for Robotics	2	1	3
BITS F343	Fuzzy Logic and Applications	3	0	3
BITS F364	Human – Computer Interaction	3	0	3
BITS F441	Robotics			3
BITS F452	Blockchain Technology	3	0	3
BITS F459	Computer Vision	3	1	4
BITS F471	Introduction to Large Language Models	3	0	3
CS F212	Database Systems	3	1	4
CS F315	Information and Communication Technologies and Development	3	0	3
CS F316	Quantum Architecture and Programming	3	0	3
CS F321	System Security	3	0	3
CS F322	Knowledge Graphs-Foundations and Applications	3	0	3
CS F407	Artificial Intelligence	3	0	3
CS F413	Internetworking Technologies	3	0	3
CS F422	Parallel Computing	3	0	3
CS F424	Software for Embedded Systems	3	1	4
CS F427	Performance Analysis of Computer Networks	3	0	3
CS F434	Data science for Healthcare			3*
CS F435	Cyber Physical Systems and Security	3	1	4
CS F437	Generative Artificial Intelligence	3	0	3
CS F446	Data Storage Technologies and Networks	3	0	3
CS G513	Network Security	3	1	4
CS G514/ SS G514	Object Oriented Analysis and Design	2	2	4
CS G527	Cloud Computing			5
CS G553	Reconfigurable Computing			5
CS G557	Distributed Computing			5*
EEE F245	Control System Laboratory	0	1	1
EEE F311	Communication Systems	3	1	4
EEE F341	Analog Electronics	3	1	4

EEE F318	FPGA Based System Design	1	2	3
EEE F411	Internet of Things	3	1	4
EEE F417	Computer based control system	3	0	3
EEE F422	Modern Control Systems	3	0	3
EEE F432	Medical Instrumentation	3	0	3
EEE F434	Digital Signal Processing	3	1	4
EEE F435	Digital Image Processing	3	0	3
EEE G512	Embedded System Design	3	1	4
EEE G513	Machine Learning for Electronics Engineers			4
IS F311	Computer Graphics	3	0	3
IS F341	Software Engineering	3	1	4
MEL G621	VLSI Design	3	2	5

ELECTRONICS AND INSTRUMENTATION ENGINEERING

CORE COURSES		**L**	**P**	**U**
INSTR F211	Electrical Machines	3	1	4
INSTR F212	Electromagnetic Theory	3	0	3
INSTR F214	Electronic Devices	3	0	3
INSTR F215	Digital Design	3	1	4
INSTR F241	Microprocessors and interfacing	3	1	4
INSTR F242	Control Systems	3	0	3
INSTR F243	Signals & Systems	3	0	3
INSTR F244	Microelectronic Circuits	3	0	3
INSTR F311	Electronic Instrumentation & Instrumentation Technology	3	1	4
INSTR F312	Transducers and Measurement Systems	3	0	3
INSTR F313	Analog & Digital VLSI Design	3	0	3
INSTR F341	Analog Electronics	3	1	4
INSTR F342	Power Electronics	3	1	4
INSTR F343	Industrial Instrumentation & Control	3	0	3

DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
BITS F312	Neural Network & Fuzzy Logic	3	0	3
BITS F415	Introduction To MEMS	3	1	4
CS F213	Object Oriented Programming	3	1	4
CS F342	Computer Architecture	3	1	4
CS F372	Operating Systems	3	0	3
CS F451	Combinatorial Mathematics	3	0	3
CS G553	Reconfigurable Computing			5
ECE F312	EM Fields and Microwave Engineering Laboratory	0	1	1
ECE F314	Electromagnetic Fields & Microwave Engineering	3	0	3
ECE F331	Microwave Engineering	3	1	4
EEE F245	Control System Laboratory	0	1	1
EEE F246	Electrical and Electronic Circuits Laboratory	0	2	2
EEE F311	Communication Systems	3	1	4
EEE F345	Power Apparatus & Networks	3	0	3
EEE F346	Data Communication Networks	2	0	2
EEE F318	FPGA Based System Design	1	2	3
EEE F411	Internet of Things	3	1	4
EEE F417	Computer Based Control System	3	0	3
EEE F419	Flexible and Stretchable Electronics	3	1	4
EEE F420	Biomedical Signal Processing	3	1	4
EEE F422	Modern Control Systems	3	0	3
EEE F423	Electronic Material Design and Simulations	1	2	3
OR	OR			
ECE F423	Electronic Material Design and Simulations	1	2	3
OR	OR			
INSTR F423	Electronic Material Design and Simulations	1	2	3
EEE F426	Fiber optics & Optoelectronics	3	0	3
EEE F427	Electric Power Utilization and Illumination	3	0	3
EEE F429	Smart Materials and Applications	3	1	4
EEE F430	Green Communications and Networks	3	0	3
EEE F431	Mobile Telecommunication Networks	3	0	3
EEE F433	Electromagnetic Fields & Waves	3	0	3
EEE F434	Digital Signal Processing	3	1	4
EEE F435	Digital Image Processing (check number and description with BITS Image Processing)	3	0	3
EEE F436	Electromagnetic Compatibility	3	1	4
EEE F437	Semiconductor Fabrication Technology	3	1	4
EEE F438	Electric Vehicles	3	0	3
EEE F472	Satellite Communication	3	0	3
EEE F474	Antenna Theory and Design	3	1	4
EEE F475	Special Electrical Machines	3	1	4
EEE F476	Switchgear and Protection	3	1	4
EEE F477	Modelling of Field-Effect Nano Devices	3	0	3
EEE F478	Power Systems Laboratory	0	2	2
EEE G512	Embedded System Design	3	1	4
EEE G513	Machine Learning for Electronics Engineers			4*
EEE G626	Hardware Software Co-Design			4
INSTR F216	Electronic Devices Simulation Laboratory	0	2	2
INSTR F413	Advanced Process Control	3	0	3
INSTR F414	Telecommunication Switching Systems & Networks	3	0	3

INSTR F415	Digital Control	3	0	3
INSTR F419	Virtual Instrumentation	3	1	4
INSTR F420	Design of Instrumentation Systems	3	0	3
INSTR F422	Instrumentation for Petrochemical Industry	3	0	3
INSTR F424	Smart Grid for Sustainable Energy	3	0	3
INSTR F428	Energy Storage Systems	3	0	3
INSTR F429	Smart Materials and Applications	3	1	4
INSTR F430	Green Communications and Networks	3	0	3
INSTR F432	Medical Instrumentation	3	0	3
INSTR F473	Wind Electrical Systems	3	0	3

ENVIRONMENTAL AND SUSTAINABILITY ENGINEERING

CORE COURSES		**L**	**P**	**U**
ENVS F211	Hydraulics and Fluid Mechanics	3	0	3
ENVS F212	Environmental Chemistry and Environmental Biotechnology	3	0	3
ENVS F221	Water and Wastewater Treatment Systems	3	0	3
ENVS F222	Transport Phenomena for Environmental Systems	3	0	3
ENVS F223	Material Science and Engineering for Sustainability	3	0	3
ENVS F224	Sampling and Analytical Techniques for Pollutants	3	0	3
ENVS F311	Environmental Modelling	3	0	3
ENVS F312	Air Pollution Control	3	0	3
ENVS F313	Digital Sustainability	3	0	3
ENVS F314	Environmental Engineering Lab	1	2	3
ENVS F321	Sustainability Policies and Measures	3	0	3
ENVS F322	Solid Waste Management	3	0	3
ENVS F232	Sustainable Urban Design and Smart Cities	3	0	3

DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
BIO F241	Ecology and Environmental Science	3	0	3
BITS F429	Nanotechnology for Renewable Energy and Environment	3	1	4
CE F412	Disaster Management	3	0	3
CE F423	Green Buildings and Energy Conservation	3	0	3
CHE F423	Membrane Science and Engineering	3	0	3
CHEM F333	Chemistry of Materials	3	0	3
CHEM F431	Sustainable Chemistry	3	0	3
	using Renewables			
ENVS F331	Climate Change Adaptation and Mitigation	3	0	3
ENVS F332	Energy Transition and Storage	3	0	3
ENVS F333	Environmental and Climate Technology Ventures	3	0	3
MF F442	Advances in Materials Science	3	0	3
MF F485	Sustainable Manufacturing	3	0	3

MANUFACTURING ENGINEERING

CORE COURSES		**L**	**P**	**U**
MF F211	Mechanics of Solids	3	0	3
MF F216	Materials Science and Engineering	2	1	3
MF F217	Machine Drawing	0	2	2
MF F218	Transport Phenomena in Manufacturing	3	1	4
MF F219	Operations Management	3	0	3
MF F220	Metrology and Quality Assurance	2	1	3
MF F221	Mechanisms and Machines	3	0	3
MF F222	Casting, Forming and Welding	3	1	4
MF F314	Design of Machine Elements	3	0	3
MF F315	Automation and Control	3	1	4
MF F316	Machining and Machine Tools	3	1	4
MF F317	Computer Aided Design and Manufacturing	2	1	3
MF F318	Non Traditional Manufacturing Processes	3	0	3
MF F319	Supply Chain Management	3	0	3
MF F320	Engineering Optimization	3	0	3

DISCIPLINE ELECTIVE COURSES				
BITS F415	Introduction To MEMS	3	1	4
ECON F411	Project Appraisal	3	0	3
ME F321	Data Mining in Mechanical Sciences	2	1	3
ME F323	Energy Storage Technologies	3	0	3
ME F340	Introduction to Sports Engg.	3	0	3
ME F416	Reverse Engineering and Rapid Prototyping	3	0	3
ME F417	Advanced Metal Forming	3	0	3
ME F419	Total Product Integration Engineering	3	0	3
ME F424	Energy Management	3	0	3
ME F425	Additive Manufacturing	3	0	3
ME F426	Industry 4.0 in Manufacturing	3	0	3
ME F432	Computer Aided Manufacturing	2	1	3
ME F436	Sustainable Energy Informatics	3	0	3
ME F443	Quality Control Assurance and Reliability	3	0	3
ME F484	Automotive Technology	3	0	3

MF F321	Procurement Management	3	0	3
MF F411	Fluid Power Systems	3	1	4
MF F412	Automotive Systems	3	0	3
MF F413	Mechanical Vibrations and Acoustics	3	0	3
MF F414	Manufacturing Excellence	3	0	3
MF F418	Lean Manufacturing	3	0	3
MF F422	Supply Chain Modelling and Empirical Analysis	3	1	4
MF F442	Advances in Materials Science	3	0	3
MF F453	Industrial Relations	3	0	3
MF F463	Maintenance and Safety	3	0	3
MF F471	Instrumentation and Control	3	0	3
MF F472	Precision Engineering	3	0	3
MF F473	Product Design and Development	3	0	3
MF F474	Product Design and Development Projects	0	0	3
MF F485	Sustainable Manufacturing	3	0	3

MATHEMATICS AND COMPUTING CORE COURSES		**L**	**P**	**U**
MAC F211	Linear Algebra and Its Applications	3	0	3
MAC F212	Object Oriented Programming	3	1	4
MAC F213	Discrete Mathematics	3	0	3
MAC F214	Elementary Real Analysis	3	0	3
MAC F241	Numerical Analysis	3	0	3
MAC F242	Data Structures & Algorithms	3	1	4
MAC F243	Numerical Optimization	3	0	3
MAC F244	Stochastic Calculus and Application to Finance	3	0	3
MAC F245	Scientific Computing Laboratory	0	1	1
MAC F311	Algebra I	3	0	3
MAC F312	Foundations of Data Science	3	0	3
MAC F313	Statistical Data Analysis	3	1	4
MAC F314	Mathematical Modelling	3	1	4
MAC F341	Design and Analysis of Algorithms	3	0	3
MAC F342	Computational Partial Differential Equations	3	1	4

DISCIPLINE ELECTIVE COURSES				
BITS F311	Image Processing	3	0	3
BITS F316	Nonlinear Dynamics and Chaos	3	0	3
BITS F386	Quantum Information and Computation	3	0	3
BITS F463	Cryptography	3	0	3
BITS F464	Machine Learning	3	0	3
CS F212	Database Systems	3	1	4
CS F402	Computational Geometry	3	0	3
CS F407	Artificial Intelligence	3	0	3
CS F415	Data Mining	3	0	3
CS F422	Parallel Computing	3	0	3
CS F425	Deep Learning	3	0	3
CS F426	Graph Mining	3	1	4
CS G513	Network Security	3	1	4
ECON F354	Derivatives and Risk Management	3	0	3
IS F311	Computer Graphics	3	0	3
MAC F411	Computation of Option Pricing Models	3	1	4
MATH F243	Graphs and Networks	3	0	3
MATH F315	Introduction to Statistical Inference	3	0	3
MATH F424	Applied Stochastic Process	3	1	4
MATH F425	Numerical Linear Algebra	3	1	4
MATH F426	Mathematical Theory of Finite Element Methods	3	1	4
MATH F428	Time Series Analysis and Forecasting	3	1	4

MECHANICAL ENGINEERING CORE COURSES		**L**	**P**	**U**
ME F211	Mechanics of Solids	3	0	3
ME F212	Fluid Mechanics	3	0	3
ME F216	Materials Science and Engineering	2	1	3
ME F217	Applied Thermodynamics	3	1	4
ME F218	Advanced Mechanics of Solids	2	0	2
ME F219	Manufacturing Processes	3	1	4
ME F220	Heat Transfer	3	1	4
ME F221	Mechanisms and Machines	3	0	3
ME F314	Design of Machine Elements	3	0	3
ME F315	Advanced Manufacturing Processes	2	1	3
ME F316	Manufacturing Management	2	0	2
ME F317	Engines, Motors, and Mobility	2	0	2
ME F318	Computer-Aided Design	1	2	3
ME F319	Vibrations and Control	3	0	3
ME F320	Engineering Optimization	3	0	3
ME F341	Prime Movers & Fluid Machines	2	1	3

DISCIPLINE ELECTIVE COURSES				
AN F311	Principles of Aerodynamics	3	0	3
AN F312	Aircraft Propulsion	3	0	3
AN F313	Flight Mechanics and Controls	3	0	3
AN F314	Introduction to Flight	3	0	3
AN F315	Aircraft Structures	3	0	3
AN F316	Orbital Mechanics	3	0	3
BITS F327	Artificial Intelligence for Robotics	2	1	3
BITS F415	Introduction to MEMS	3	1	4

BITS F441	Robotics	3	0	3
BITS F451	Autonomous Mobile Robotics	3	0	3
BITS F462	Renewable Energy	3	0	3
DE G513	Tribology	3	2	5
DE G514	Fracture Mechanics	3	2	5
DE G531	Product Design	3	2	5
ECE F242	Control Systems	3	0	3
ECON F411	Project Appraisal	3	0	3
EEE F242	Control Systems	3	0	3
INSTR F242	Control Systems	3	0	3
MATH F313	Numerical Analysis	3	0	3
ME F321	Data Mining in Mechanical Sciences	2	1	3
ME F323	Energy Storage Technologies	3	0	3
ME F324	Cell and Tissue Biomechanics	3	0	3
ME F325	Fundamentals of Soft Matter	3	1	4
ME F340	Introduction to Sports Engineering	3	0	3
ME F411	Fluid Power Systems	3	1	4
ME F413	Nonlinear Vibrations	3	0	3
ME F414	Fuel Cell Science and Technology			3*
ME F415	Gas Dynamics	3	0	3
ME F416	Reverse Engineering and Rapid Prototyping	3	0	3
ME F417	Advanced Metal Forming	3	0	3
ME F418	Rocket and Spacecraft Propulsion	3	0	3
ME F419	Total Product Integration Engg.	3	0	3
ME F420	Power Plant Engineering	3	0	3
ME F423	Microfluidics and Applications			4*
ME F424	Energy Management	3	0	3
ME F425	Additive Manufacturing	3	0	3
ME F426	Industry 4.0 in Manufacturing	3	0	3
ME F427	Continuum Mechanics	3	1	4
ME F428	Smart Materials	3	1	4
ME F429	Micro-Nanoscale Heat Transport	3	1	4
ME F430	Fluid-Structure Interactions	3	0	3
ME F432	Computer Aided manufacturing	2	1	3
ME F433	Solar Thermal Process Engineering	3	1	4
ME F434	Digital Twins in Mechanical Engineering	3	1	4
ME F435	Shape Memory Alloys: Fundamentals and Applications	2	1	3
ME F436	Sustainable Energy Informatics	3	0	3
ME F437	Thermo-fluidic of Food Process Engineering	3	0	3
ME F438	Micro and Nano Energy Harvesting Technology	3	1	4
ME F441	Automotive Vehicles	3	0	3
ME F443	Quality Control, Assurance and Reliability	3	0	3
ME F451	Mechanical Equipment Design	3	0	3
ME F452	Composite Materials & Design	3	0	3
ME F461	Refrigeration and Air conditioning	3	0	3
ME F472	Precision Engineering	3	0	3
ME F482	Combustion	3	0	3
ME F483	Wind Energy	3	0	3
ME F484	Automotive Technology	3	0	3
ME F485	Numerical Techniques for Fluid Flow and Heat Transfer	3	0	3
ME G511	Mechanism and Robotics	3	2	5
ME G512	Finite Element Methods	3	2	5
ME G514	Turbomachinery	3	2	5
ME G515	Computational Fluid Dynamics	3	2	5
ME G533	Conduction and Radiation Heat Transfer	3	2	5
ME G534	Convective Heat and Mass Transfer	3	2	5
ME G544	Multibody Dynamics	3	2	5
ME G614	Topology Optimization of Machine Components	3	2	5
MF F311	Mechatronics and Automation	2	1	3
MF F321	Procurement Management	3	0	3
MF F418	Lean Manufacturing	3	0	3
MF F421	Supply chain management			4
MF F422	Supply Chain Modelling and Empirical Analysis	3	1	4
MF F485	Sustainable Manufacturing	3	0	3
MST G522	Advanced Composites	3	2	5

MECHANICAL ENGINEERING WITH SPECIALIZATION IN AEROSPACE

CORE COURSES		**L**	**P**	**U**
AN F211	Fluid Mechanics and Machines	3	1	4
AN F221	Production Techniques	2	1	3
AN F311	Principles of Aerodynamics	3	0	3
AN F312	Aircraft Propulsion	3	0	3
AN F315	Aircraft Structures	3	0	3
AN F321	Aircraft Design	1	1	2
ME F211	Mechanics of Solids	3	0	3
ME F216	Materials Science and Engineering	2	1	3
ME F217	Applied Thermodynamics	3	1	4
ME F220	Heat Transfer	3	1	4
ME F221	Mechanisms and Machines	3	0	3
ME F314	Design of Machine Elements	3	0	3
ME F316	Manufacturing Management	2	0	2
ME F317	Engines, Motors, and Mobility	2	0	2
ME F318	Computer-Aided Design	1	2	3
ME F319	Vibrations and Control	3	0	3

DISCIPLINE ELECTIVE COURSES				
AN F313	Flight Mechanics and Control	3	0	3
AN F314	Introduction to Flight	3	0	3
AN F316	Orbital Mechanics	3	0	3
ME F212	Fluid Mechanics	3	0	3
ME F218	Advanced Mechanics of Solids	2	0	2
ME F219	Manufacturing Processes	3	1	4
ME F315	Advanced Manufacturing Processes	2	1	3
ME F320	Engineering Optimization	3	0	3
ME F341	Prime Movers & Fluid Machines	2	1	3
ME F415 or ME F418	Gas Dynamics or Rocket and Spacecraft Propulsion	3	0	3
ME F452	Composite Materials and Design	3	0	3
PHY F319	Spacecraft Systems	3	0	3

PHARMACY CORE COURSES		L	P	U
BITS F219	Process Engineering	2	1	3
PHA F211	Pharmaceutical Analysis	2	1	3
PHA F214	Anatomy, Physiology & Hygiene	2	1	3
PHA F215*	Introduction to Molecular Biology and Immunology	3	0	3
PHA F216	Pharmaceutical Formulations I	2	1	3
PHA F217	Pharmaceutical Microbiology	2	1	3
PHA F241	Pharmaceutical Chemistry	2	1	3
PHA F242	Biological Chemistry	2	1	3
PHA F243	Industrial Pharmacy	2	1	3
PHA F244	Physical Pharmacy	2	1	3
PHA F311	Pharmacology I	2	1	3
PHA F312	Medicinal Chemistry I	2	1	3
PHA F313	Instrumental Methods of Analysis	2	1	4
PHA F315	Pharmaceutical Formulations II	2	1	3
PHA F341	Pharmacology II	2	1	3
PHA F342	Medicinal Chemistry II	2	1	3
PHA F343	Forensic Pharmacy	2	-	2
PHA F344	Natural Drugs	2	1	3

*** To be offered to B.Pharm. students admitted in 2014 onwards in place of PHA F243**

DISCIPLINE ELECTIVE COURSES		L	P	U
BITS F467	Bioethics and Biosafety	3	0	3
MATH F212	Optimization	3	0	3
PHA F316	Pharmaceutical Regulatory Science	3	0	3
PHA F317	Safety Pharmacology and Toxicology	3	0	3
PHA F413	Pharmaceutical Management and Quality Control	3	0	3
PHA F414	Biopharmaceutics	3	0	3
PHA F415	Pathophysiology	3	0	3
PHA F416	Chemistry of Synthetic Drugs	3	0	3
PHA F417	Pharmacoeconomics	3	0	3
PHA F418	Biopharmaceutics and Pharmacokinetics	3	0	3
PHA F419	Herbal Drug Technology	3	0	3
PHA F422	Cosmetic Science	2	1	3
PHA F432	Hospital Pharmacy	3	0	3
PHA F441	Biochemical Engineering	3	0	3
PHA F442	Applied Pharmaceutical Chemistry	3	0	3
PHA F461	Phytochemistry	2	1	3
PHA G546	Pharmaceutical Biostatistics	3	0	3

BIOLOGICAL SCIENCES CORE COURSES		L	P	U
BIO F211	Biological Chemistry	3	0	3
BIO F212	Microbiology	3	1	4
BIO F213	Cell Biology	3	0	3
BIO F214	Integrated Biology	3	0	3
BIO F215	Biophysics	3	0	3
BIO F241	Ecology & Environmental Science	3	0	3
BIO F242	Introduction to Bioinformatics	3	0	3
BIO F243	Genetics	3	0	3
BIO F244	Instrumental Methods of Analysis	1	3	4
BIO F311	Recombinant DNA Technology	3	0	3
BIO F312	Plant Physiology	3	0	3
BIO F313	Animal Physiology	3	0	3
BIO F341	Developmental Biology	3	0	3
BIO F342	Immunology	3	0	3

DISCIPLINE ELECTIVE COURSES		L	P	U
BIO F216	Water, Sanitation and Solid Waste Management	3	0	3
BIO F217	Laboratory for Water, Sanitation and Solid waste Management	1	2	3
BIO F231	Biology Project Laboratory			3
BIO F314	Conservation Biology	2	1	3
BIO F315	Applied Nutrition and Nutraceuticals	2	0	2
BIO F352	Cell and Tissue Culture Technology	3	1	4
BIO F411	Laboratory	0	3	3
BIO F413	Molecular Biology of Cell	3	0	3
BIO F417	Biomolecular Modelling	3	0	3
BIO F418	Genetic Engineering Techniques	1	3	4
BIO F419	Molecular Evolution	3	0	3
BIO F421	Enzymology	3	0	3

BIO F431	Reproductive Physiology	3	0	3
BIO F441	Biochemical Engineering	3	0	3
BIO F451	Bioprocess Technology	3	0	3
BIO G512	Molecular Mechanism of Gene Expression	3	2	5
BIO G513	Microbial and Fermentation Technology	3	2	5
BIO G515	Stem Cell and Regenerative Biology	3	1	4
BIO G522	Interferon Technology	3	1	4
BIO G523	Advanced and Applied Microbiology	3	2	5
BIO G524	Animal Cell Technology	3	2	5
BIO G525	Environmental Biotechnology & Waste Mgnt	3	2	5
BIO G526	Cancer Biology	3	2	5
BIO G528	Developmental Neuroscience	3	2	5
BIO G544	Bioremediation and biometallurgy	5	0	5
BIO G545	Molecular Parasitology and Vector Biology			5
BIO G561	Advances in Recombinant DNA Technology	3	2	5
BIO G570	Recent Developments in Biology	1	0	1
BIO G612	Human Genetics	3	2	5
BIO G631	Membrane and Liposome Technology	3	1	4
BIO G632	Transgenic Technology	3	2	5
BIO G642	Experimental Techniques		4	4
BIO G643	Plant Biotechnology	3	2	5
BIO G651	Protein and Enzyme Bioengineering	3	2	5
BIO G661	Gene Toxicology	3	1	4
BIO G671	Bioconversion Technology	3	2	5
BIOT F345	Proteomics	3	0	3
BIOT F346	Genomics	3	0	3
BIOT F347	Immunotechnology	3	0	3
BIOT F416	Introduction to Pharmaceutical Biotechnology	3	0	3
BIOT F422	Nanobiotechnology	3	0	3
BIOT F424	Food Biotechnology	3	0	3
BITS F418	Introduction to Biomedical Engineering	3	1	4
BITS F467	Bioethics and Biosafety	3	0	3
CHEM F212	Organic Chemistry I	3	0	3
CHEM F213	Physical Chemistry II	3	0	3
MATH F212	Optimization	3	0	3
PHA G529	Neuropharmacology			5

CHEMISTRY CORE COURSES		**L**	**P**	**U**
CHEM F211	Physical Chemistry I	3	0	3
CHEM F212	Organic Chemistry I	3	0	3
CHEM F213	Physical Chemistry II	3	0	3
CHEM F214	Inorganic Chemistry I	3	0	3
CHEM F241	Inorganic Chemistry II	3	0	3
CHEM F242	Chemical Experimentation I	0	3	3
CHEM F243	Organic Chemistry II	3	0	3
CHEM F244	Physical Chemistry III	3	0	3
CHEM F311	Organic Chemistry III	3	0	3
CHEM F312	Physical Chemistry IV	3	0	3
CHEM F313	Instrumental Methods of Analysis	3	1	4
CHEM F341	Chemical Experimentation II	0	4	4
CHEM F342	Organic Chemistry IV	3	0	3
CHEM F343	Inorganic Chemistry III	3	0	3
PHY F212	Electromagnetic Theory I	3	0	3

DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
CHEM F223	Colloid and Surface Chemistry	3	0	3
CHEM F320	Introductory Computational Chemistry Laboratory	0	4	2
CHEM F323	Biophysical Chemistry	3	0	3
CHEM F324	Numerical Methods in Chemistry	3	3	4
CHEM F325	Polymer Chemistry	3	0	3
CHEM F326	Solid State Chemistry	3	0	3
CHEM F327	Electrochemistry: Fundamentals and Applications	3	0	3
CHEM F328	Supramolecular Chemistry	3	0	3
CHEM F329	Analytical Chemistry	3	1	4
CHEM F330	Photophysical Chemistry	3	1	4
CHEM F333	Chemistry of Materials	3	0	3
CHEM F334	Magnetic Resonance	3	0	3
CHEM F335	Organic Chemistry and Drug Design	3	0	3
CHEM F336	Nanochemistry	3	1	4
CHEM F337	Green Chemistry and Catalysis	3	0	3
CHEM F412	Photochemistry and Laser Spectroscopy	3	0	3
CHEM F413	Electron Correlation In Atoms And Molecules	3	1	4
CHEM F414	Bio and Chemical Sensors	3	0	3
CHEM F415	Frontiers in Organic Synthesis	3	0	3
CHEM F416	Applied Crystallography	3	0	3
CHEM F422	Statistical Thermodynamics	3	0	3
CHEM F423	Astrochemistry	3	0	3
CHEM F430	Atmospheric Chemistry	3	0	3
CHEM F431	Sustainable Chemistry using Renewables	3	0	3
CHEM G513	Advanced Nuclear and Radio-chemistry			5
CHEM G521	Environmental Chemistry			5
CHEM G541	Chemical Applications of Group Theory			5

CHEM G555	Chemistry of Life Processes	4
CHEM G556	Catalysis	4
CHEM G557	Solid Phase Synthesis and Combinatorial Chemistry	4
CHEM G558	Electronic Structure Theory	5
CHEM G559	Bioinorganic Chemistry	4
CHEM G561	Heterocyclic Chemistry	5
CHEM G562	Solid State Chemistry	4
CHEM G563	Advanced Statistical Mechanics	5

ECONOMICS

CORE COURSES		**L**	**P**	**U**
ECON F211	Principles of Economics	3	0	3
ECON F212	Fundamentals of Finance and Accounts	3	0	3
ECON F213	Mathematical and Statistical Methods	3	0	3
ECON F214	Economic Environment of Business	3	0	3
ECON F241	Econometric Methods	3	0	3
ECON F242	Microeconomics	3	0	3
ECON F243	Macroeconomics	3	0	3
ECON F244	Economics of Growth and Development	3	0	3
ECON F311	International Economics	3	0	3
ECON F312	Money, Banking and Financial Markets	3	0	3
ECON F313	Issues in Economic Development	3	0	3
ECON F341	Public Finance Theory and Policy	3	0	3
ECON F342	Applied Econometrics	3	0	3
ECON F343	Economic Analysis of Public Policy	3	0	3

DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
ECON F215	Computational Methods for Economics	3	0	3
BITS F314	Game Theory and It's Applications	3	0	3
ECON F315	Financial Management	3	0	3
ECON F314	Industrial Economics	3	0	3
ECON F345	Behavioral Economics	3	0	3
ECON F351	Indian Economic Development	3	0	3
ECON F352	Management of Banks and Financial Institutions	3	0	3
ECON F353	Energy Economics and Policy	3	0	3
ECON F354	Derivatives and Risk Management	3	0	3
ECON F355	Business Analysis and Valuation	3	0	3
ECON F356	Strategic Financial Management	3	0	3
ECON F357	Management Control System	3	0	3
ECON F411	Project Appraisal	3	0	3
ECON F412	Security Analysis and Portfolio Management	3	0	3
ECON F413	Financial Engineering	3	0	3
ECON F414	Creating and Leading Entrepreneurial Organizations	3	0	3
ECON F415	New Venture Creation	3	0	3
ECON F417	Risk Management and Insurance	3	0	3
ECON F418	Quantitative Analysis of International Trade	3	0	3
ECON F419	Advanced Microeconomics	3	0	3
ECON F420	Applied Macroeconometrics	3	0	3
ECON F422	Functions and Working of Stock Exchanges	3	0	3
ECON F434	International Business	3	0	3
ECON F435	Marketing Research	3	0	3
ECON F471	Resources and Environmental Economics	3	0	3
FIN F314	Investment Banking and Financial Services	3	0	3
FIN F414	Financial Risk Analytics and Management	3	0	3
MATH F212	Optimization	3	0	3
MATH F242	Operations Research	3	0	3
MATH F424	Applied Stochastic Process	3	1	4

MATHEMATICS

CORE COURSES		**L**	**P**	**U**
MATH F212	Optimization	3	0	3
MATH F213	Discrete Mathematics	3	0	3
MATH F214	Elementary Real Analysis	3	0	3
MATH F215	Algebra I	3	0	3
MATH F241	Mathematical Methods	3	0	3
MATH F242	Operations Research	3	0	3
MATH F243	Graphs and Networks	3	0	3
MATH F244	Measure & Integration	3	0	3
MATH F311	Introduction to Topology	3	0	3
MATH F312	Ordinary Differential Equations	3	0	3
MATH F313	Numerical Analysis	3	0	3
MATH F341	Introduction to Functional Analysis	3	0	3
MATH F342	Differential Geometry	3	0	3
MATH F343	Partial Differential Equations	3	0	3

DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
BITS F314	Game Theory and Its Applications	3	0	3
BITS F316	Nonlinear Dynamics and Chaos	3	0	3
BITS F343	Fuzzy Logic and Applications	3	0	3
BITS F368	Quantum Information and Computing	3	0	3
BITS F463	Cryptography	3	0	3

CS F211	Data Structures and Algorithms	3	1	4
or BITS F232	**or** Foundations of Data Structures and Algorithms	3	1	4
CS F364	Design and Analysis of Algorithms	3	0	3
MAC F211	Linear Algebra and Its Applications	3	0	3
MAC F244	Stochastic Calculus and Application to Finance	3	0	3
MAC F313	Statistical Data Analysis	3	1	4
MAC F411	Computation of Option Pricing Models	3	1	4
MATH F231	Number Theory	3	0	3
MATH F314	Algebra II	3	0	3
MATH F315	Introduction to Statistical Inference	3	0	3
MATH F316	Matrix Theory and Linear Estimation	3	1	4
MATH F317	Sampling Theory	3	1	4
MATH F353	Statistical Inference and Applications	3	0	3
MATH F354	Complex Analysis	3	0	3
MATH F378	Advanced Probability Theory	3	0	3
MATH F420	Mathematical Modeling	3	0	4
MATH F421	Combinatorial Mathematics	3	0	3
MATH F422	Numerical Methodology for Partial Differential Equations	3	1	4
MATH F423	Introduction to Algebraic Topology	3	0	3
MATH F424	Applied Stochastic Process	3	1	4
MATH F425	Numerical Linear Algebra	3	1	4
MATH F426	Mathematical Theory of Finite Element Methods	3	1	4
MATH F427	Statistical Simulation and Data Analysis	3	1	4
MATH F428	Time Series Analysis and Forecasting	3	1	4
MATH F429	Modern Theory of Partial Differential Equations			3
MATH F431	Distribution Theory	3	0	3
MATH F432	Applied Statistical Methods	3	0	3
MATH F441	Discrete Mathematical Structures	3	0	3
MATH F444	Numerical Solutions of Ordinary Differential Equations	3	0	3
MATH F445	Mathematical Fluid Dynamics	3	0	3
MATH F456	Cosmology	3	0	3
MATH F471	Nonlinear Optimization	3	0	3
MATH F481	Commutative Algebra	3	0	3
MATH F492	Wavelet analysis and applications	3	1	4

PHYSICS

CORE COURSES		**L**	**P**	**U**
PHY F211	Classical Mechanics	3	1	4
PHY F212	Electromagnetic Theory I	3	0	3
PHY F213	Optics	3	0	3
PHY F214	Electricity, Magnetism & Optics Laboratory	0	2	2
PHY F241	Electromagnetic Theory II	3	1	4
PHY F242	Quantum Mechanics I	3	0	3
PHY F243	Mathematical Methods of Physics	3	0	3
PHY F244	Modern Physics Laboratory	0	2	2
PHY F311	Quantum Mechanics II	3	0	3
PHY F312	Statistical Mechanics	3	0	3
PHY F313	Computational Physics	3	0	3
PHY F341	Solid State Physics	3	0	3
PHY F342	Atomic & Molecular Physics	3	0	3
PHY F343	Nuclear & Particle Physics	3	0	3
PHY F344	Advanced Physics Laboratory	0	3	3
DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
BIO F215	Biophysics	3	0	3
BITS F316	Nonlinear Dynamics and Chaos	3	0	3
BITS F317	Theoretical Neuroscience	3	0	3
BITS F386	Quantum Information and Computation	3	0	3
BITS F416	Introduction to Nanoscience	3	0	3
BITS F417	Microfluidics & its Applications			4
BITS F446	Pattern Recognition			3
EEE F426	Fibre Optics & Optoelectronics	3	0	3
MATH F424	Applied Stochastic Processes	3	1	4
MATH F456	Cosmology	3	0	3
PHY F215	Introduction to Astronomy & Astrophysics	3	0	3
PHY F315	Theory of Relativity	3	0	3
PHY F316	Musical Acoustics	3	0	3
PHY F317	Introduction to Radio Astronomy	3	0	3
PHY F318	Atoms and Photons	3	0	3
PHY F346	Laser Science and Technology	3	0	3
PHY F378	Plasma Physics and its Applications	3	0	3
PHY F379	Thin Film Technology	3	0	3
PHY F412	Introduction To Quantum Field Theory	3	1	4
PHY F413	Particle Physics	3	1	4
PHY F414	Physics of Advanced Materials	3	1	4
PHY F415	General Theory of Relativity and Cosmology	3	1	4
PHY F416	Soft Condensed Matter Physics	3	1	4

PHY F417	Experimental Methods of Physics	3	1	4
PHY F418	Lasers and Applications	3	1	4
PHY F419	Advanced Solid State Physics	3	1	4
PHY F420	Quantum Optics	3	1	4
PHY F421	Advanced Quantum Mechanics	3	1	4
PHY F422	Group Theory and Applications	3	1	4
PHY F423	Special Topics in Statistical Mechanics	3	1	4
PHY F424	Advanced Electrodynamics	3	1	4
PHY F425	Advanced Mathematical Methods of Physics	3	1	4
PHY F426	Physics of Semiconductor Devices	3	1	4
PHY F427	Atmospheric Physics	3	0	3
PHY F428	Quantum Information Theory	3	0	3
PHY F431	Geometrical Methods in Physics	3	0	3
PHY F432	Classical Theory of Fields: A Symmetry Perspective			4*
PHY F433	Topics in Nonlinear Optics	3	0	3
PHY F434	Foundations of Quantum Mechanics	3	0	3
PHY F435	Advanced Computational Physics	3	0	3
PHY F438	Fundamentals of Quantum Field Theory	3	0	3
PHY G512	Advanced Quantum Field Theory	3	0	3

PHYSICS WITH SPECIALIZATION IN SPACE SCIENCE AND TECHNOLOGY

CORE COURSES		**L**	**P**	**U**
PHY F211	Classical Mechanics	3	1	4
PHY F212	Electromagnetic Theory I	3	0	3
PHY F213	Optics	3	0	3
PHY F214	Electricity, Magnetism &Optics Laboratory	0	2	2
PHY F241	Electromagnetic Theory II	3	1	4
PHY F242	Quantum Mechanics I	3	0	3
PHY F243	Mathematical Methods of Physics	3	0	3
PHY F244	Modern Physics Laboratory	0	2	2
PHY F311	Quantum Mechanics II	3	0	3
PHY F312	Statistical Mechanics	3	0	3
PHY F313	Computational Physics	3	0	3
PHY F341	Solid State Physics	3	0	3
PHY F342	Atomic & Molecular Physics	3	0	3
PHY F343	Nuclear & Particle Physics	3	0	3
PHY F344	Advanced Physics Laboratory	0	3	3

DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
CE F433	Remote Sensing and Image processing	3	1	4
ECE F472	Satellite Communication	3	0	3
MATH F456	Cosmology	3	0	3
ME F418	Rocket and Spacecraft Propulsion	3	0	3
PHY F215	Introduction to Astronomy and Astrophysics	3	0	3
PHY F317	Introduction to Radio Astronomy	3	0	3
PHY F319	Spacecraft Systems	3	0	3
PHY F320	Space Physics	3	0	3
PHY F415	General Theory of Relativity and Cosmology	3	1	4
PHY F436	Space Science Instrumentation	3	1	4
PHY F437	Scientific Computing and Data Analysis	3	1	4

ROBOTICS AND INDUSTRIAL AUTOMATION

CORE COURSES		**L**	**P**	**U**
RIA F211	Materials and Mechanics for Robotics			3*
RIA F212	Introduction to Mechanisms & Robotics			3*
RIA F213	Foundations of Data Structures and Algorithms	3	1	4
RIA F214	Sensors, Actuators and Signal Conditioning	2	1	3
RIA F241	Artificial Intelligence for Robotics	2	1	3
RIA F242	Control Systems	3	0	3
RIA F243	Signals and Systems	3	0	3
RIA F244	Embedded Systems for Robotics	2	1	3
RIA F311	Fluid Power Systems	3	1	4
RIA F312	Robot Dynamics & Control			3*
RIA F313	Computer Vision	3	1	4
RIA F341	IoT for Industrial Automation			3*
RIA F342	Computer-Aided Design	1	2	3
RIA F343	Autonomous Mobile Robotics	3	0	3
RIA F344	Industrial Automation & Control	2	1	3

DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
BITS F311	Image Processing	3	0	3
BITS F312	Neural Networks and Fuzzy Logic	3	0	3
BITS F434	Fuzzy Logic and Applications	3	0	3
BITS F364	Human Computer Interaction	3	0	3
BITS F372	Data Communications and Networks	3	0	3
BITS F415	Introduction to MEMS	3	1	4
BITS F442	Remote Sensing and Image Processing			3
BITS F454	Bio-Inspired Intelligence: Algorithms and Applications	3	0	3
BITS F464	Machine Learning	3	0	3
BITS G541	User Interfaces	1	3	4
CS F213	Object Oriented Programming	3	1	4
OR	OR			
IS F213	Object Oriented Programming	3	1	4
OR	OR			
MAC F312	Object Oriented Programming	3	1	4

CS F314	Software Development for Portable Devices	2	1	3
CS F320	Foundations of Data Science	3	0	3
OR	OR			
MAC F312	Foundations of Data Science	3	0	3
CS F407	Artificial Intelligence	3	0	3
CS F415	Data Mining	3	0	3
CS F424	Software for Embedded Systems	3	1	4
CS F425	Deep Learning	3	0	3
CS F436	Cyber Physical Systems and Security	3	1	4
CS F444	Real–Time Systems	3	0	3
ECE F343	Communication Networks	3	0	3
ECE F434	Digital Signal Processing	3	1	4
ECOM F321	Real Time Operating Systems	3	1	4
EEE F241	Microprocessors and Interfacing	3	1	4
OR	OR			
ECE F241	Microprocessors and Interfacing	3	1	4
OR	OR			
ECOM F241	Microprocessors and Interfacing	3	1	4
OR	OR			
INSTR F241	Microprocessors and Interfacing	3	1	4
EEE F313	Analog & Digital VLSI Design	3	0	3
OR	OR			
ECE F313	Analog & Digital VLSI Design	3	0	3
OR	OR			
ECOM F313	Analog & Digital VLSI Design	3	0	3
OR	OR			
INSTR F313	Analog & Digital VLSI Design	3	0	3
EEE F411	Internet of Things	3	1	4
EEE F417	Computer Based Control Systems	3	0	3
EEE F418	Modern Communication Technologies	3	0	3
OR	OR			
ECE F418	Modern Communication Technologies	3	0	3
EEE F422	Modern Control Systems	3	0	3
EEE F428	Energy Storage Systems	3	0	3
OR	OR			
ECE F428	Energy Storage Systems	3	0	3
OR	OR			
INSTR F428	Energy Storage Systems	3	0	3
EEE F432	Medical Instrumentation	3	0	3
INSTR F413	Advanced Process Control	3	0	3
INSTR F419	Virtual Instrumentation	3	1	4
ME F314	Design of Machine Elements	3	0	3
ME F320	Engineering Optimization	3	0	3
ME F321	Data Mining in Mechanical Sciences	2	1	3
ME F424	Energy Management	3	0	3
ME F425	Additive Manufacturing	3	0	3
ME F426	Industry 4.0 in Manufacturing	3	0	3
ME F432	Computer Aided Manufacturing	3	0	3
ME F434	Digital Twins in Mechanical Engineering	3	1	4
ME F439	Autonomous Aerial Vehicles	3	0	3
ME G511	Mechanisms & Robotics	2	3	5
ME G512	Finite Element Methods			5
ME F418	Lean Manufacturing	3	0	3
RIA F321	Underwater Robotics	3	0	3
RIA F322	Autotronics and Vehicle Intelligence	3	0	3
RIA F323	AR/VR in Robotics			3*
RIA F371	Capstone Project-1			3
RIA F372	Capstone Project-2			3

GENERAL STUDIES – COMMUNICATION AND MEDIA STUDIES STREAM

CORE COURSES		**L**	**P**	**U**
GS F221	Business Communication	3	0	3
GS F222	Language Lab Practice	0	3	3
GS F223	Introduction to Mass Communication	3	0	3
GS F224	Print and Audio Visual Advertising	2	1	3
GS F241	Creative Writing	2	1	3
GS F243	Current Affairs	3	0	3
GS F244	Reporting and Writing for Media	3	0	3
GS F245	Effective Public Speaking	2	1	3
GS F321	Mass Media Content and Design	2	1	3
GS F322	Critical Analysis of Literature and Cinema	3	0	3
GS F342	Computer Mediated Communication	3	0	3
GS F343	Short Film and Video Production	2	1	3

DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
BITS F385	Introduction to Gender Studies	3	0	3
GS F211	Modern Political Concepts	3	0	3
GS F212	Environment, Development & Climate Change	3	0	3
GS F231	Dynamics of Social Change	3	0	3
GS F232	Introductory Psychology	3	0	3
GS F234	Development Economics	3	0	3
GS F242	Cultural Studies	3	0	3
GS F311	Introduction to Conflict Management	3	0	3
GS F312	Applied Philosophy	3	0	3
GS F313	Marxian Thoughts	3	0	3
GS F325	Journalism	3	0	3
GS F326	Creative Thinking	2	1	3
GS F327	Selected Reading	3	0	3
GS F331	Techniques in Social Research	3	0	3
GS F333	Public Administration	3	0	3
GS F334	Global Business Technology & Knowledge Sharing	3	0	3
GS F344	Copywriting	2	0	2

HSS F227	Cross Cultural Skills	3	0	3
HSS F232	Introduction to Development Studies	3	0	3
HSS F247	Social Informatics	3	0	3
HSS F315	Society, Business, and Politics	3	0	3
HSS F317	Introduction to Globalization	3	0	3
HSS F319	Lighting for Theatre and Films			2
HSS F323	Organizational Psychology	3	0	3
HSS F328	Human Resource Development	3	0	3
HSS F341	Performance Design	1	2	3
HSS F343	Professional Ethics	3	0	3
HSS F346	International Relations	3	0	3
HSS F384	Indian National Movement	3	0	3

GENERAL STUDIES – DEVELOPMENT STUDIES STREAM

CORE COURSES		**L**	**P**	**U**
ECON F211	Principles of Economics	3	0	3
GS F211	Modern Political Concepts	3	0	3
GS F212	Environment, Development & Climate Change	3	0	3
GS F213	Development Theories	3	0	3
GS F231	Dynamics of Social Change	3	0	3
GS F232	Introductory Psychology	3	0	3
GS F233	Public Policy	3	0	3
GS F234	Development Economics	3	0	3
GS F311	Introduction to Conflict Management	3	0	3
GS F312	Applied Philosophy	3	0	3
GS F331	Techniques in Social Research	3	0	3
GS F332	Contemporary India	3	0	3
GS F333	Public Administration	3	0	3
GS F334	Global Business Technology & Knowledge Sharing	3	0	3

DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
BITS F214	Science, Technology and Modernity	3	0	3
BITS F385	Introduction to Gender Studies	3	0	3
BITS F399	Humanistic Theories of Science and Technology	3	0	3
GS F313	Marxian Thoughts	3	0	3
HSS F233	Main Trends in Indian History	3	0	3
HSS F234	Main Currents of Modern History	3	0	3
HSS F235	Introductory Philosophy	3	0	3
HSS F236	Symbolic Logic	3	0	3
HSS F247	Social Informatics	3	0	3
HSS F312	Bureaucracy	3	0	3
HSS F315	Society, Business, and Politics	3	0	3
HSS F343	Professional Ethics	3	0	3
HSS F344	Heritage of India	3	0	3
HSS F345	Gandhian Thoughts	3	0	3
HSS F346	International Relations	3	0	3
HSS F384	Indian National Movement	3	0	3

SEMICONDUCTOR AND NANOSCIENCE

CORE COURSES		**L**	**P**	**U**
SNS F211	Fundamentals of Nanoscience and Nanotechnology	3	0	3
SNS F212	Quantum Mechanics	3	0	3
SNS F213	Material Science and Engineering	3	0	3
SNS F214	Fundamentals of Solid State Physics	3	0	3
SNS F241	Chemistry at Nanoscale	3	0	3
SNS F242	Electronic Devices	3	0	3
SNS F243	Materials & Nanoscale Characterization	3	1	4
SNS F244	Nanomaterials Synthesis & Characterization Laboratory	0	3	3
SNS F311	Semiconductor Thin Film Technology	3	0	3
SNS F312	Molecular Simulations	2	1	3
SNS F313	Instrumental Methods of Analysis	3	1	4
SNS F341	Principles of Nanobiotechnology	3	0	3
SNS F342	Principles of Nanophotonics and Devices	3	0	3
SNS F343	Advanced Nanofabrication	0	3	3

DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
BIO F417	Biomolecular Modelling	3	0	3
BIO G510	Application of Computers and Statistics in Biology			5
BIOT F416	Introduction to Pharmaceutical Biotechnology	3	0	3
BIOT F423	Drug design and delivery	3	0	3
BITS F415	Introduction to MEMS	3	1	4
BITS F418	Introduction to Biomedical Engineering	3	1	4
BITS F429	Nanotechnology for Renewable energy and Environment	3	1	4
BITS F467	Bioethics and Biosafety	3	0	3
CHE F415	Molecular and Statistical Thermodynamics	3	0	3
CHE F433	Corrosion Engineering	3	0	3
CHE F498	Colloids and Interface Engineering			3
OR	OR			
CHEM F223	Colloid and Surface Chemistry	3	0	3

CHEM F327	Electrochemistry: Fundamentals and Applications	3	0	3
CHEM F328	Supramolecular Chemistry	3	0	3
CHEM F333	Chemistry of Materials	3	0	3
CHEM F414	Bio and Chemical Sensors	3	0	3
ECE F423	Electronic Material Design and Simulations Laboratory	1	2	3
EEE F419	Flexible and Stretchable Electronics	3	1	4
EEE F477	Modelling of Field-Effect Nano Devices	3	0	3
ME F423	Microfluidics and Application			4*
ME F452	Composite Materials and Design	3	0	3
MEL G514	Nanoelectronic Memories and Technology	3	2	5
MF F442	Advances in Materials Sciences	3	0	3
MST F333	Introduction to Biomaterials	3	0	3
MST F337	Materials for Energy Applications	3	0	3
MST F338	Metals and Alloys	3	0	3
MST F339	Polymer Materials	3	0	3
MST G522	Advanced Composites	3	2	5
PHY F414	Physics of Advanced Materials	3	1	4
PHY F419	Advanced Solid State Physics	3	1	4
SNS F315	Nanotechnology & Nanomaterials in Sensors	3	0	3
SNS F345	Physics and Modelling of Semiconductor Devices	2	1	3

CORE COURSES		**L**	**P**	**U**
BBA F121	Business Ethics and Corporate Social Responsibility	3	0	3
BBA F211	Financial and Management Accounting	3	0	3
BBA F221	Human Resource Management	3	0	3
BBA F222	Business Law and Compliance	3	0	3
BBA F311	Design Thinking	2	2	4
BBA F321	Digital Enterprises	3	0	3
BITS F330	Negotiation Skills and Techniques			3*
ECON F414	Creating and Leading Entrepreneurial Organization	3	0	3
BITS F428	Essentials of Strategic Management	3	0	3
ECON F434	International Business	3	0	3
MF F219	Operations Management	3	0	3
MGTS F311	Marketing	3	0	3
MGTS F314	Essentials of Financial Management	3	0	3
MGTS F351	Organizational Behaviour	3	0	3

DISCIPLINE ELECTIVE COURSES		**L**	**P**	**U**
Pool – I: Finance				
BBA F341	Behavioral Finance	3	0	3
BBA F342	Fintech	3	0	3
BBA F343	Mergers and Acquisitions	3	0	3
BBA F411	Forensic Accounting	3	0	3
BBA F412	Islamic Finance	3	0	3
BITS F493	Business Analysis and Valuation	3	0	3
ECON F312	Money, Banking and Financial Markets	3	0	3
ECON F352	Management of Banks and Financial Institutions	3	0	3
ECON F356	Strategic Financial Management	3	0	3
ECON F413	Financial Engineering	3	0	3
FIN F243	Functions & Working of Stock Exchanges	3	0	3
FIN F311	Derivatives & Risk Management	3	0	3
FIN F313	Security Analysis & Portfolio Management	3	0	3
FIN F414	Financial Risk Analytics & Management	3	0	3
Pool – II: Marketing				
BBA F344	Marketing Channels	3	0	3
BBA F355	International Marketing Strategies	3	0	3
BBA F345	Introduction to Consumer Behaviour	3	0	3
BBA F346	Marketing Research	3	0	3
BBA F413	Sales and Distribution Management	3	0	3
BBA F414	Introduction to Services Marketing	3	0	3
BBA F415	Customer Relationship Management	3	0	3
BBA F416	Fundamentals of Retailing	3	0	3
BITS F427	Digital Marketing	3	0	3
MGTS F313	Product and Brand Management	3	0	3
MGTS F433	Advertising and Sales Promotion	3	0	3
Pool – III: Business Analytics				
BBA F345	Financial Analytics	3	0	3
BBA F346	Predictive Analytics	3	0	3
BBA F347	Marketing Analytics	3	0	3
BBA F348	People Analytics	3	0	3
BBA F417	Prescriptive Analytics	3	0	3
BBA F418	Pricing Analytics			3*
BBA F419	Time Series Analysis and Forecasting	3	0	3
BITS F455	Analytics for Supply Chain			3*
Pool – IV: Organizational Behaviour and Human Resource Management				
BBA F349	Training and Development	3	0	3
BBA F350	Performance Management	3	0	3
BBA F351	Compensation Management	3	0	3
BBA F352	Strategic Human Resource Management	3	0	3
BBA F353	Organizational Change and Development	3	0	3
BBA F420	Work, Health and Safety	3	0	3

BBA F421	Leadership Skills	3	0	3
BBA F422	International Human Resource Management	3	0	3
BBA F423	Talent Management and Development	3	0	3
BBA F424	Strategic Leadership	3	0	3
BITS G517	Cross Cultural Management	3	0	3
GS F311	Introduction to Conflict Management	3	0	3
MF F453	Industrial Relations	3	0	3

Pool – V: Operations and Supply Chain

BBA F425	Introduction to Logistics Management	3	0	3
BITS F490	Project Management			4
ECON F344	Models in Operations Management	3	0	3
ECON F411	Project Appraisal	3	0	3
ME F412	Production Planning and Control	3	0	3
ME F443	Quality Control Assurance and Reliability	3	0	3
MF F319	Supply Chain Management	3	0	3
MF F321	Procurement Management	3	0	3
MF F418	Lean Manufacturing	3	0	3
MF F422	Supply Chain Modelling and Empirical Analysis	3	1	4
MF F485	Sustainable manufacturing	3	0	3

Project Type Courses

In addition to discipline electives mentioned above, the following project type courses are also being offered by the departments for each of their respective programmes. These courses may be taken by the students to meet the discipline elective requirements.

XXX F266	Study Project	3
XXX F366	Laboratory Project	3
XXX F367	Laboratory Project	3
XXX F376	Design Project	3
XXX F377	Design Project	3
XXX F491	Special Project	3

where XXX indicates the Degree programme. For example, CHE F266 Study Project is intended for a student of B.E. Chemical Engineering.

A student may avail a maximum of 3 Project courses to meet the Discipline Electives Requirement under the head of (Discipline) Electives with the following limitations:

(a) All of these Project courses should be

  (i) within the Discipline (for which the degree is being awarded) or

  (ii) from an allied Discipline if so specified by the Department offering the degree

(b) The projects may be chosen from under these sub-heads.

  (i) Study Projects (maximum of 1)

  (ii) Laboratory (maximum of 2)

  (iii) Design Projects (maximum of 2)

  (iv) Special Projects (maximum of 1)

A student may avail a maximum of 3 Project courses (under any of the heads mentioned above offered by any discipline as an Open Elective. However, in total a student may avail at most 5 Project courses against Electives slots in any category.

Pool of Humanities courses for first degree programmes:

The following is the list of courses from which Humanities Electives can be taken by the students in different first degree programs to meet the general institutional requirement of eight units under the Humanities elective category:

Course No.	Course Title	L	P	U
BITS F214	Science, Technology and Modernity	3	0	3
BITS F226	Soft Skills for Professionals	3	0	3
BITS F239	Evolution of Architecture	2	0	2
BITS F328	Innovation Ecosystem and the Future of Work			4*
BITS F385	Introduction to Gender Studies	3	0	3
BITS F399	Humanistic Theories of Science and Technology	3	0	3
BITS F419	Management of Cross Cultural Engineering Teams	3	0	3
BITS F458	Professional Practice and Ethics	3	0	3
GS F211	Modern Political Concepts	3	0	3
GS F212	Environment, Development & Climate Change	3	0	3
GS F213	Development Theories	3	0	3
GS F221	Business Communication	3	0	3
GS F222	Language Lab Practice	0	3	3
GS F223	Introduction to Mass Communication	3	0	3
GS F224	Print and Audio-Visual Advertisement	2	1	3

Course No.	Course Title	L	P	U
GS F231	Dynamics of Social Change	3	0	3
GS F232	Introductory Psychology	3	0	3
GS F233	Public Policy	3	0	3
GS F234	Development Economics	3	0	3
GS F241	Creative Writing			3*
GS F242	Cultural Studies	3	0	3
GS F243	Current Affairs	3	0	3
GS F244	Reporting and Writing for Media	2	1	3
GS F245	Effective Public Speaking	2	1	3
GS F311	Introduction to Conflict Management	3	0	3
GS F312	Applied Philosophy	3	0	3
GS F313	Marxian Thoughts	3	0	3
GS F321	Mass Media Content and Design	2	1	3
GS F322	Critical Analysis of Literature and Cinema	3	0	3
GS F325	Journalism	3	0	3
GS F326	Creative Thinking	2	1	3
GS F327	Selected Reading	3	0	3
GS F331	Techniques in Social Research	3	0	3
GS F332	Contemporary India	3	0	3
GS F333	Public Administration	3	0	3
GS F334	Global Business, Technology and Knowledge Sharing	3	0	3
GS F343	Short Film and Video Production	2	1	3
GS F344	Copywriting	2	0	2
HSS F211	Introduction to Arabic	3	0	3
HSS F221	Readings from Drama	3	0	3
HSS F222	Linguistics	3	0	3
HSS F223	Appreciation of Indian Music	3	0	3
HSS F224	English Skills for Academic	3	0	3
HSS F226	Postmodernism	3	0	3
HSS F227	Cross Cultural Skills	3	0	3
HSS F228	Phonetics & Spoken English	3	0	3
HSS F229	Introduction to Western Music	3	0	3
HSS F232	Introduction to Development Studies	3	0	3
HSS F233	Main Trends in Indian History	3	0	3
HSS F234	Main Currents of Modern History	3	0	3
HSS F235	Introductory Philosophy	3	0	3
HSS F236	Symbolic Logic	3	0	3
HSS F237	Contemporary Indian English Fiction	3	0	3
HSS F238	Sports and Society	3	0	3
HSS F244	Crime and New Media	3	0	3
HSS F245	Gender, Science and Technology	3	0	3
HSS F246	Philosophy of Nāgārjuna	3	0	3
HSS F247	Social Informatics	3	0	3
HSS F248	Introduction to Disability Studies	3	0	3
HSS F249	Politics in India	3	0	3
HSS F250	Comics and Visual Culture	3	0	3
HSS F251	Introduction to Discourse and Conversational Analysis	2	1	3
HSS F252	International Law	3	0	3
HSS F253	Upcycling Theory	3	0	3
HSS F254	Food and Society	3	0	3
HSS F266	Study Project			3
HSS F311	Introduction to Videogame Studies	3	0	3
HSS F313	Introduction to Contemporary Arts	3	0	3
HSS F314	Maritime Studies and Blue Economy	3	0	3
HSS F315	Society, Business, and Politics	3	0	3
HSS F316	Popular Literature and Culture of South Asia	3	0	3
HSS F317	Introduction to Globalization	3	0	3
HSS F318	Introduction to Anthropology	3	0	3
HSS F319	Lighting for Theatre and Films			2*
HSS F323	Organizational Psychology	3	0	3
HSS F325	Cinematic Adaptation	3	0	3
HSS F326	Humanities and Design	2	1	3
HSS F327	Contemporary Drama	3	0	3
HSS F328	Human Resource Development	3	0	3
HSS F329	Musicology – An – Introduction	3	0	3
HSS F330	Appreciation of Art	3	0	3
HSS F331	Sankara's Thoughts	3	0	3

Course No.	Course Title	L	P	U
HSS F332	Cinematic Art			3*
HSS F333	Comparative Religion	3	0	3
HSS F334	Srimad Bhagavad Gita	3	0	3
HSS F335	Literary Criticism	3	0	3
HSS F336	Modern Fiction	3	0	3
HSS F337	English Literary Forms and Movements	3	0	3
HSS F338	Comparative Indian Literature	3	0	3
HSS F339	Theatre Art Acting and Production	3	0	3
HSS F340	Post Colonial Literatures	3	0	3
HSS F341	Performance Design	1	2	3
HSS F342	Advanced Communicative English	3	0	3
HSS F343	Professional Ethics	3	0	3
HSS F344	Heritage of India	3	0	3
HSS F345	Gandhian Thoughts	3	0	3
HSS F346	International Relations	3	0	3
HSS F347	Introduction to Carnatic Music	3	0	3
HSS F348	Introduction to Hindustani Music	3	0	3
HSS F349	Ecocriticism	3	0	3
HSS F350	Human Rights: History, Theory & Practice	3	0	3
HSS F351	Social and Political Ecology	3	0	3
HSS F352	Technology, Work and Society	3	0	3
HSS F353	Philosophy of Aesthetics	3	0	3
HSS F354	Introduction to Islamic Economy	3	0	3
HSS F355	Dictatorship, Democracy & Development	3	0	3
HSS F356	Social Movements and Protest Politics	3	0	3
HSS F364	Political Economy of Gulf Cooperation Council States	3	0	3
HSS F365	Science of Sustainable Happiness	3	0	3
HSS F368	Asian Cinemas and Cultures	3	0	3
HSS F369	Caste and Gender in India	3	0	3
HSS F371	Cities-Life, Issues and Conflicts	3	0	3
HSS F372	Introduction to Social Psychology	3	0	3
HSS F373	Shakespeare and Popular Culture	3	0	3
HSS F374	Urban Modernity and the Renewal of Paris	3	0	3
HSS F375	Business and Politics in Colonial and Post Colonial India: a historical approach	3	0	3
HSS F378	Cinemas of India: From Past to Present	3	0	3
HSS F380	Performance Studies	3	0	3
HSS F381	Environmental Sustainability Ventures	3	0	3
HSS F382	Housing: programs, policies and practices	3	0	3
HSS F383	Introduction to Museum Studies	3	0	3
HSS F384	Indian National Movement	3	0	3
HSS F385	Mental Health Literacy for Youth	3	0	3
HSS F386	Ocean Humanities	3	0	3
HSS F387	History of Transport and Mobility in Modern Asia	3	0	3
HSS F388	Gandhi in Literature	3	0	3
HSS F389	Introduction to Data and Society	3	0	3
HSS F390	Science and the Humanities: the Two Culture Debate	3	0	3
HSS F391	Acoustic Humanities	2	1	3
HSS F392	Crisis Communication	3	0	3
HSS F393	Introduction to Metacognition	3	0	3
HSS F394	The Stud of Aging & Soc across cultures	3	0	3
HSS F395	Chinese Politics	3	0	3
HSS F396	Indian Psychology	3	0	3
HSS F399	Introduction to American Literature	3	0	3
SANS F111	Sanskrit	3	0	3

It may be noted that a student cannot count a course (or its equivalent) of his/her own discipline (s) as a humanities elective even if it is listed in this pool of humanities electives.

Other Courses

BIO F231	Biology Project Laboratory	3
BITS C483	Indian Wisdom for Modern Management	3 0 3
BITS F211	Introduction to IPR	1
BITS F212	Introduction to Human Rights	1
BITS F213	Introduction to Environmental studies	1

BITS F215	Applications of Bio-Medical Instrumentation Techniques in Healthcare	2 0 2
BITS F217	Environment, Development and Climate Change	3 0 3
*BITS F225	Environmental Studies	3 0 3
BITS F311	Image Processing	3 0 3
BITS F320	Managerial Skills	2*
BITS F321	Legal and Economic Environment of Business	4*
BITS F330	Negotiation Skills and Techniques	3*
BITS F333	Project on Organisational Aspects	3
BITS F334	Project on Organisational Aspects	3
BITS F372	Data Communications and Networks	3 0 3
BITS F381	TIC Projects	3
BITS F382	Reading Course	3
BITS F383	TIC Projects	3
BITS F398	Creative Multimedia	2 2 3
BITS F414	Introduction to Bioinformatics	3 0 3
BITS F416	Introduction to Nanoscience	3 0 3
BITS F417	Micro Fluidics and its Application	4*
BITS F428	Essentials of Strategic Management	3 0 3
BITS F431	Flexible Manufacturing Systems	3 2 3
BITS F441	Robotics	3
BITS F442	Remote Sensing and Image Processing	3
BITS F444	Artificial Intelligence	3
BITS F445	Neural Networks and Applications	3 0 3
BITS F446	Pattern Recognition	3
BITS F447	Multimedia Computing	3 0 3
BITS F448	Retail Management Systems	3 0 3
BITS F449	Financial Engineering	3 0 3
BITS F461	Software Engineering	3
BITS F462	Renewable Energy	3 0 3
BITS F468	New Venture Creation	3 0 3
BITS F469	Financing Infrastructure Projects	3 0 3
BITS F488	Services Management Systems	3 0 3
BITS F490	Project Management	4
BITS F493	Business Analysis and Valuation	3 0 3
BITS F494	Environmental Impact Assessment	3 1 4
BITS G516	Introduction to Business Sustainability	3 0 3
BITS G517	Cross Cultural Management	3 0 3
MGTS F351	Organisational Behaviour	3 0 3
MGTS F433	Advertising and Sales Promotion	3 0 3
PHY F221	Modern Physics	3 0 3
PHY F345	Quantum Mechanics for Engineers	3 0 3

***[Students completing this course will be awarded a non-letter grade (GOOD or POOR)]**

List of Audit Type Courses

BITS N101T	PHY FITNESS HEALTH & WEL	1*
BITS N301T	PERSONALITY INTEGRATION LEADERSHIP ORIENTATION AND TEAMWORK	
CHI N101T	BEGINNING CHINESE	3 0 3
FRE N101T	BEGINNING FRENCH	3*
FRE N102T	TECHNICAL FRENCH	3 0 3
GER N101T	BEGINNING GERMAN	3 0 3
GER N102T	TECHNICAL GERMAN	3 0 3
HSS N201T	INTRO TO PHOTOGRAPHY	3 0 3
HSS N202T	IND C DANCE BHARATNATYAM	3 0 3
HSS N203T	BASIC ARABIC	3 0 3
HSS N301T	ELEMENTS OF DANCE	1 1 2
JAP N101T	BEGINNING JAPANESE	3 0 3
MUSIC N103T	INDIAN CLASS MUSIC VOC I	3*
MUSIC N104T	INDIA CLASS MUSIC VOC II	3*
MUSIC N105T	WESTERN CLASS MUSIC I	3*
MUSIC N106T	WESTERN CLASS MUSIC II	3*
MUSIC N111T	HIND CLASS MUSIC (INST)I	3*
MUSIC N112T	HIND CLASS MUSIC(INST)II	3*
MUSIC N113T	INDIA CLASS MUSIC(INST)I	3*
MUSIC N114T	INDIA CLASS MUSIC(INST)II	3*
MUSIC N203T	INDIA CLASS MUSIC(VOC) III	3*
MUSIC N204T	IND CLASS MUSIC(VOC) IV	3*
MUSIC N205T	WESTERN CLASS MUSIC III	3*
MUSIC N206T	WESTERN CLASS MUSIC IV	3*
MUSIC N213T	IND CLASS MUSIC (INST) III	3*
MUSIC N214T	IND CLASS MUSIC(INST)IV	3*
MUSIC N303T	ADV IND MUSIC PRA(VOCAL)	-
MUSIC N313T	ADV IND MUSIC PRA (INST)	-
RUS N101T	BEGINNING RUSSIAN	3 0 3
RUS N102T	TECHNICAL RUSSIAN	3 0 3